

FutureProof EPM

Income from Property Equity — Technical Architecture

A comprehensive guide to the platform architecture, agents, workflows, and compliance systems that power monthly income distribution for equity-partnered homeowners.

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1. EPM Overview

FutureProof's Income from Property Equity solution is a fund-based income product that transforms home equity into sustainable monthly income. The model connects four stakeholders through a sophisticated distribution system powered by monthly rebalancing and performance monitoring.

The Four Stakeholders

Stakeholder	Role	Capital Flow
Customer	Homeowner who partners equity for mortgage	Receives monthly distributions from fund returns
Investment Partner	Manages invested capital	Generates returns; manages fund performance and rebalancing
Lender	Provides mortgage product & customer relationship	Retains margin on monthly distributions (0.75%-2%)
Wholesale Funder	Supplies capital to lenders via funding lines	Receives wholesale margin on deployed capital (0.2%-3.6%)

Monthly Distribution Flow

Fund Returns: Investment Partner's holdings generate returns aggregated and tracked monthly (on average 10% p.a. over 30 years).

Monthly Distribution: System calculates monthly payout amounts based on:

- Customer's equity percentage (typically 2% of home value)
- Fund performance month-to-date
- Interest holiday status (if triggered)

Amount to Customer: Customer receives their share of monthly returns as income.

Lender Margin: Deducted from distribution (0.75%-2% depending on tier).

Wholesale Margin: Received by Wholesale Funder (0.2%-3.6% depending on structure).

FutureProof Margin: Platform takes 0.25% for ongoing management.

2. Customer Journey

PHASE 1: ORIGINATION (Weeks 1–4)

- **Customer applies:** Provides property details, equity percentage, income expectations via portal
- **Deal Agent processes:** KYC/AML checks, property valuation, risk tier assignment, document generation
- **e-Signature:** Customer reviews and signs loan documents, investment agreement
- **Funding confirmation:** Lender & Wholesale Funder approve, capital transferred, account activated

PHASE 2: ONGOING SERVICING (Months 1–30+)

- **Monthly distributions:** Distribution Agent calculates returns every month
- **Performance monitoring:** Investment Agent tracks fund performance, generates rolling metrics on dashboard
- **Interest holidays:** If performance drops, holiday flag triggered automatically; status visible on statements
- **Statements:** Show distribution amount, margins paid, fund performance, holiday status
- **Account management:** Customer can request early exit; lender reviews within 5 business days

PHASE 3: EXIT (Early or Maturity)

- **Customer-initiated:** Request early repayment; Deal Agent calculates exit position
- **Lender approval:** Reviews exit request, confirms terms
- **Settlement:** Distribution Agent processes final payment, accounts reconciled, loan closed
- **Archive:** All transaction history preserved 7 years; final audit report completed

3. Four-Agent Architecture

Agent	Primary Role	Key Workflows	Triggers
Deal Agent	Customer acquisition & onboarding	KYC/AML checks, property valuation, risk assessment	New deal, compliance clearance, fund confirmation
Distribution Agent	Monthly returns & payment	Calculate distributions, execute payments	Monthly statements, compliance updates, holiday changes
Investment Agent	Fund monitoring & holidays	Monitor performance, calculate rolling metrics	Monthly updates, threshold breaches, recovery
Compliance Agent	Fraud detection & enforcement	Real-time checks, fraud scoring, violation tracking	Regulatory updates, fraud scores, rule violations

Architecture: Rails services → Sidekiq (4 workers) → Kafka (event streaming) → OpenTelemetry (tracing) → State machines → Twilio Flex (call center) → Circuit breakers (sync/async)

4. Interest Holiday Mechanism

Key Concept: Interest holidays FLAG market conditions—they do NOT suspend payments. Customer continues receiving distributions; interest accumulates in the account instead of being paid out.

Parameter	Value	Meaning
Entry Threshold	0.90 (90%)	Fund drops to 90% of starting value → holiday flagged
Exit Threshold	1.458 (145.8%)	Fund recovers to 145.8% → holiday removed
Accumulated Interest	Ongoing	During holiday, interest compounds in account
Monitoring	Monthly	Investment Agent checks thresholds monthly

Timeline Example: Months 1-10 normal (distributions) → Month 11 fund drops to 88%, holiday flagged → Months 12-18 flag active (customer still gets distributions, interest compounds) → Month 19 fund recovers to 145.8%, flag removed → Normal operations

5. Compliance & Fraud Detection

Rule	Validation	Failure Action
KYC/AML	Identity verified before onboarding	Application rejected
Responsible Lending	Fund amount appropriate for profile	Risk tier adjusted
Distribution Accuracy	Calculations & margins verified	Distribution held; audit initiated
Fund Performance Audit	Returns match investment partner	Discrepancy flagged
Holiday Logic	Thresholds correctly applied	Manual review by ops

Real-Time Monitoring: Account anomalies (unusual logins, rapid device changes) • Unusual transactions (multiple same-day withdrawals) • Data inconsistencies (fund value mismatches) • Regulatory violations (undisclosed conflicts)

6. Technology Stack

Backend & Processing

- Rails 8.1 (Ruby 3.4.8)
- Sidekiq 7.x (4 worker pool)
- ruby-kafka (event streaming)
- Puma (3 threads, auto-scaling)

Data & Persistence

- PostgreSQL 16 (primary + replica)
- Redis 7 (cache, sessions)
- Kafka 3.6 (3-replica topics)
- PgBouncer (pooling)

Customer Communication

- SendGrid (email)
- Twilio (SMS + voice)
- Twilio Flex (call center)
- DocuSign (e-signature)

Frontend & Infra

- Hotwire + Stimulus JS
- AWS S3 (docs)
- Fly.io (deployment)
- GitHub Actions (CI/CD)

Monitoring & Security

- Sentry (errors)
- CloudWatch (logs)
- Grafana (dashboards)
- PagerDuty (escalation)
- HTTPS/TLS + CSP

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